SHIP TYPES 1.1 · 10 MIN

Classification of Ship Types

Main types of ships: **cargo carrying** (bulk carriers, dry cargo and passenger ships, tankers) · **work vessels** (fishing, offshore, dredging, tugs, service craft) · **non-merchant ships** (naval, yachts, training) · inland, non-propelled and non-ship units

Session W1.1 · Not in the Evaluation Test

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Ten Minutes on a Ship Type, Every Lecture Session

Week	Wed 13:00–15:00	Wed 15:00–16:30	Fri 13:30–15:30	Fri 15:30–17:00
Week 1 **4–6 Nov**	**W1.1 Classification** IMO, EU, StatCode 5	**W1.2 What limits a ship** Deadweight, volume, size	**W1.3 Container ships** Feeder to ULCV, securing	**W1.4 Bulk carriers** Handysize to Capesize
Week 2 **11–13 Nov**	**W2.1 Guest lecture** No ship type	**W2.2 Guest lecture** No ship type	**W2.3 General cargo** MPP, heavy-lift, reefers	**W2.4 Ro-Ro ships** Con-Ro, PCC, PCTC
Week 3 **18–20 Nov**	**W3.1 Crude tankers** Aframax to VLCC, shuttle	**W3.2 Product tankers** MR, LR, chemical, bunker	**W3.3 Gas carriers** LPG, LNG, ammonia, CO2	**W3.4 Ferries** Ro-Pax, fast ferries
Week 4 **25–27 Nov**	**W4.1 Cruise ships** Cruise, expedition, yachts	**W4.2 Drilling units** Jack-up, semi-sub, drillship	**W4.3 Production units** FPSO, FSO, FLNG, spar, TLP	**W4.4 Evaluation Test** No ship type
Week 5 **2–4 Dec**	**W5.1 Support vessels** PSV, AHTS	**W5.2 Offshore construction** Pipe-layers, crane, DSV	**W5.3 Wind installation** WTIV, heavy-lift jack-ups	**W5.4 Floating wind** Foundations, cable-layers
Week 6 **9–11 Dec**	**W6.1 Wind service** SOV, CTV	**W6.2 Surface combatants** Frigates, corvettes	**W6.3 Project clinic** No ship type	**W6.4 Project clinic** No ship type
Week 7 **16–18 Dec**	**W7.1 Patrol and MCM** OPV, coastguard, uncrewed	**W7.2 Amphibious, support** Replenishment, submarines	**W7.3 Presentations** No ship type	**W7.4 Presentations** No ship type

Classification 2 · Merchant 9 · Oil and gas 4 · Wind 3 · Naval 3 · No ship type 7

Merchant Ships: the Focus of the Course

1 Dry bulk carriers

Handysize to Capesize bulk carriers, ore carriers, self-unloaders

W1.4

2 General and special cargo

General cargo and multi-purpose ships, heavy-lift and semi-submersible ships, reefers, livestock carriers

W2.3

3 Container ships

Feeder to ultra-large, gearless and geared

W1.3

4 Ro-Ro and vehicle carriers

Ro-Ro, Con-Ro, pure car and truck carriers (PCC, PCTC)

W2.4

5 Oil and chemical tankers

Crude (Aframax to VLCC), shuttle, product, chemical and bunker tankers

W3.1 · W3.2

6 Gas carriers

LPG, LNG, ammonia and CO2 carriers; liquid hydrogen next

W3.3

7 Passenger ships and yachts

Ro-Pax and fast ferries, cruise ships, yachts

W3.4 · W4.1

Our ships: the reference ship is in group 5; the Azores Express and Feeder are in group 3.

Energy at Sea, Naval Vessels and the Rest

8 Offshore oil and gas

Drilling units (semi-submersible, jack-up, drill-ship) · production and storage (FPSO, FSO, FLNG, spar, TLP) · support vessels (PSV, AHTS) · construction (pipe-layers, crane vessels, DSV)

W4.2 · W4.3 · W5.1 · W5.2

9 Offshore wind

Installation vessels (WTIV, heavy-lift jack-ups) · floating foundations and cable-layers · service vessels (SOV, CTV)

W5.3 · W5.4 · W6.1

10 Naval and government vessels

Frigates, corvettes, destroyers · patrol, coastguard and mine countermeasure vessels, uncrewed vessels · amphibious, replenishment and support ships, submarines

W6.2 · W7.1 · W7.2

11 Tugs, dredgers and service vessels

Tugs, icebreakers, dredgers, pilot vessels, buoy tenders, oil-recovery, fire-fighting, research and survey, training ships

Today only

12 Fishing vessels

Trawlers, purse seiners, longliners, fish carriers

Today only

In total: 9 sessions on merchant ships, 4 on offshore oil and gas, 3 on offshore wind, 3 on naval vessels.

“There are no universally applicable definitions of ship types.”

IMO, Safety Regulations, imo.org

Despite the lack of cross-framework consistency, specific names and definitions are used rigorously inside each framework. IMO conventions, EU MRV and FuelEU Maritime, classification societies and commercial data providers each have their own taxonomy. Each was designed for a different purpose (regulatory compliance, commercial data, technical class rules, operational training) and each contributes something the others miss.

THE THREE KEY FRAMEWORKS

IMO conventions

Regulatory compliance

SOLAS, MARPOL and the codes; MARPOL Annex VI sets the types for EEDI, EEXI and CII

EU MRV, ETS and FuelEU

EU reporting and compliance

One list of ship types, each with its cargo measure, reused by all three

StatCode 5

Commercial data and statistics

S&P Global’s five-level coding, used by ship databases and statistics offices

SHIP TYPES 1.1 · STATCODE 5

Six Families of Ships, Coded in Five Levels

One ship, five levels

Level 1 · Family
Cargo carrying

Level 2 · Group
Dry cargo/passenger

Level 3 · Sub-group
Container

Level 4 · Ship type
Container ship

Level 5 · Detailed type
Container ship (fully cellular)

Cargo carrying

Bulk carriers · Dry cargo/passenger · Tankers

Our two ships and the reference ship

Work vessel

Fishing · Offshore · Miscellaneous (dredging, research, towing, other activities)

Non merchant

Naval and naval auxiliaries · Yachts · Sail training ships

Non seagoing merchant

Inland waterway cargo ships, tankers, passenger ships, tugs and dredgers

Non propelled

Barges and pontoons

Non ship structure

Platforms, floating docks, buoys, submersibles, hovercraft

The World Fleet in 2025: Capacity and Growth

1 January 2025 · **112,501 ships** of 100 GT and above · **2.44 billion dwt**, 3.4 % more than a year earlier

Type	Share of world deadweight	Share	Growth 2024
Bulk carriers		42.5 %	+3.0 %
Oil tankers		27.5 %	+0.7 %
Container ships		**14.9 %**	**+9.7 %**
Liquefied gas carriers		4.1 %	+7.0 %
Offshore supply		3.7 %	+2.0 %
General cargo		3.6 %	+1.9 %
Chemical tankers		2.2 %	+3.3 %
Other and unknown		1.1 %	+4.7 %
Ferries and passenger ships		0.4 %	+1.5 %

Container ships grew fastest, by **9.7 %** in one year. The fleet is also getting older: **12.6 years** on average by tonnage, **22.2 years** by number of ships.

Where the Baseline Ships Sit

Criterion	A · Azores Express	B · Azores Feeder
StatCode 5 type	Container Ship (Fully Cellular), gearless	Container Ship (Fully Cellular), own crane
Size	1,300 TEU · feeder class	130 TEU · small feeder
Operating area	Open sea: Lisbon–Azores, weekly	Short sea: inter-island loops
Hull	Conventional container ship	Hull of the group’s choice
Machinery	Dual-fuel engine, single shaft, PTI	Two electric motors; generator sets for dry-dock trips and emergencies
Energy source	HFO and LNG	Swappable battery containers

Same StatCode 5 ship type (Cargo Carrying › Dry Cargo/Passenger › Container), very different ships: 0.2 Ship Design Project, slides 4 and 5, has the owner’s requirements of each. **Next: Activity 1.1.**

Next session, Ship types 1.2: what limits the ship, its deadweight, its volume or one of its dimensions.

ACTIVITY 1.1 · INDIVIDUAL · 15 MIN · NOT GRADED

Find Each Ship in StatCode 5

Open the StatCode 5 Explorer on your phone (course site, W1.1), alone or with a neighbour. For each ship, write its five levels, from family to detailed type.

#	Ship	Your answer
1	The reference ship: a 37,600 t oil/chemical tanker	Family › … › detailed type
2	A ferry carrying cars and passengers between the Azores islands	Family › … › detailed type
3	A passenger ferry that only sails on a river	Family › … › detailed type
4	A service operation vessel for an offshore wind farm	Family › … › detailed type
5	A ship of your choice	Family › … › detailed type